Human genome and medical biotechnology

Module: SC3303

By

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Lecture 03:

Human Genome Project and its influence on medical biotechnology

Human genome project

- The Human Genome Project has transformed biology through its integrated big science approach to deciphering a reference human genome sequence along with the complete sequences of key model organisms.

- The development of novel technologies and analytical tools, and how it brought the expertise of engineers, computer scientists and mathematicians together with biologists.

- The genome sequences of microbes, plants and animals have revolutionized many fields of science, including microbiology, virology, infectious disease and plant biology.

A short history:

- The idea of sequencing the human genome was first proposed in the USA committee appointed by the US national research council. It refers to the international 13-year effort, formally begun in October 1990 and completed in 2003, to discover all estimated 20,000 to 25000 human genes.

- Another goal of projects is to determine the complete sequence of the 3 billion DNA subunits. The first genetic map was completed in September 1994 while the physical map with 30,000 or more (sequence-tagged sies) STSs was completed in October 1998.

- The sequencing of 99% of portions of gene-rich regions was completed in April 2003. Meanwhile, identification and characterization of disease-causing genes, sequencing of other model organisms and other uncompleted tasks were done in the same year.

Aims and objectives of HGP:

The core values or objectives of the present projects are to tailor a physical or genetic map of the whole human as well as mice genome.

To sequence both the genome and other genomes like yeast and microbes in order to use it as a test run for human genome sequencing.

Overall, two broader objectives of it are:

1. Creating a physical map of the human genome
2. To sequence the whole genome of around 3 billion base pairs.

Other objectives are,

To make the information available for all researchers thereby developing a computational system to store, transfer, process and retrieve the data across the world.

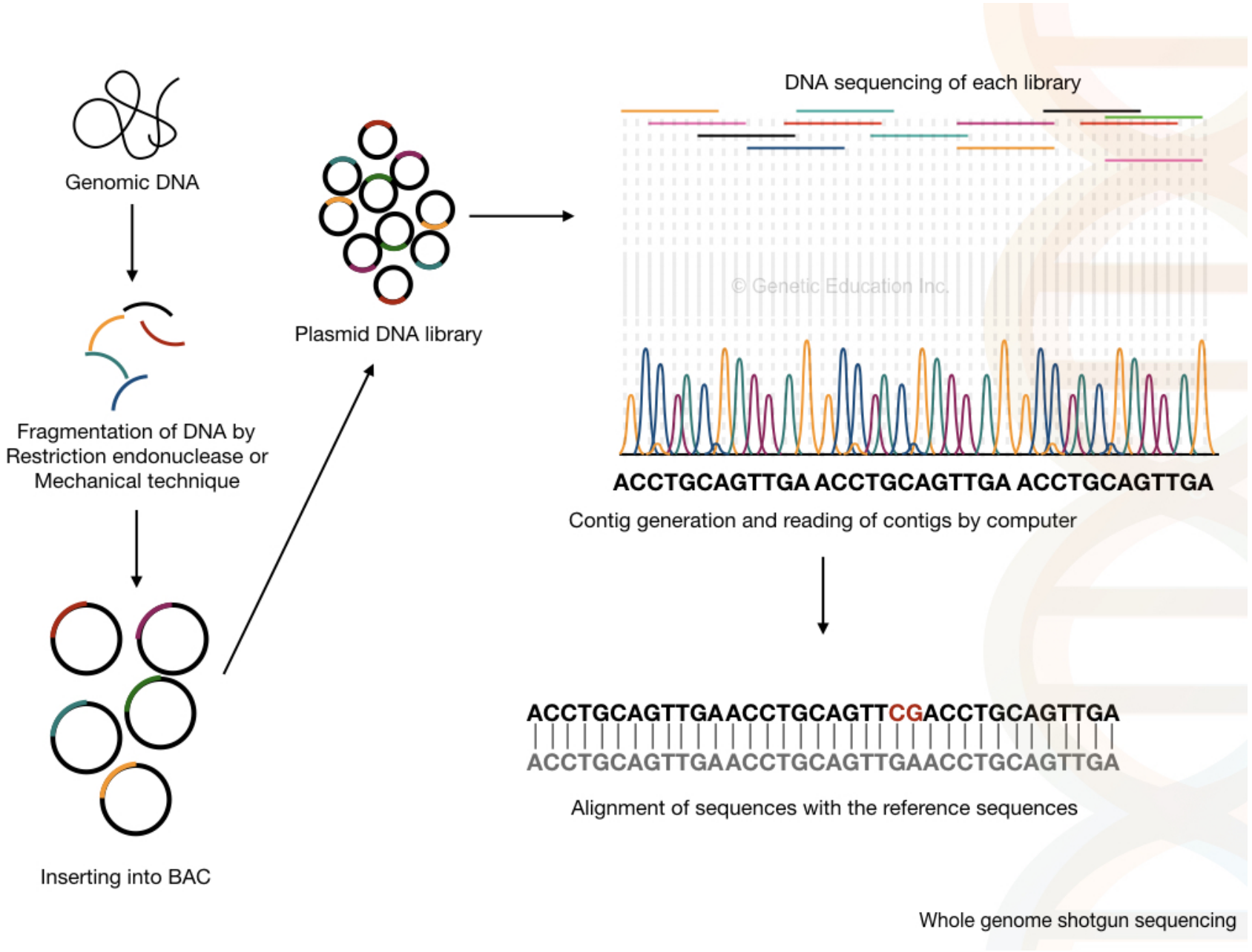
- To identify all the disease-causing genes.
- To understand the function of genes
- To map and tag genes on chromosomes.
- To develop tools to process and analyze data.

Aim and Objective

Human Genome Project

- To Sequence the whole genome with 3billion bps
- Creating a physical Map of the Human genome
- To identify all the Disease causing genes
- To understand the Function of Genes
- To make the informations Available for all Researchers
- To develop tools For processing and Analysing data

Techniques used in HGP:



Human genome project

- Moreover, deeper knowledge of human sequence variation has begun to alter the practice of medicine.

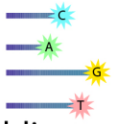
- The Human Genome Project has inspired subsequent large-scale data acquisition initiatives such as the
    - **International HapMap Project,**
    - **1000 Genomes, and**
    - **The Cancer Genome Atlas**,

- as well as the recently announced **Human Brain Project** and the emerging
    - **Human Proteome Project**.

DNA sequencing

Sanger sequencing

uses **dideoxy nucleotides** (chain terminators)

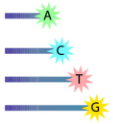


by adding a mix of normal and terminator you can get pieces of different lengths where you know the last letter

illumina

sequencing by synthesis

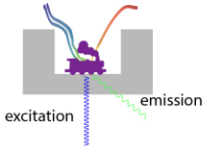
reversible-terminator nucleotides



read fluorescence after each letter is added, then remove fluorophore and allow more adding

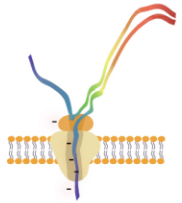
PacBio

Single Molecule Real Time Sequencing (SMRT)



read fluorescence as letter is added

NanoPore



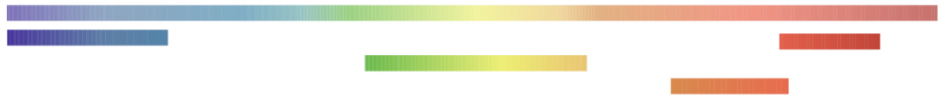
read changes in current as letters pass through pore

average read length

accuracy

combining methods

longer reads are often used as a scaffold for aligning shorter, more accurate reads - great for figuring out where repetitive sequences go



Effort required to sequence a human genome

A

2000
1st Generation
(Sanger sequencing)

Scientists – Hundreds

Machines - Hundreds

Cost $3 billion

Time - 10 years

B

2006-2010
2nd Generation
("Next Generation")

Scientists – 1-2

Machines - 1

Cost $5-10,000

Time - 2 weeks

C

2010-2015
3rd Generation
("Next-Next Generation")

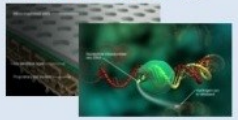
Helicos BioSciences Corporation

PACIFIC BIOSCIENCES

Oxford NANOPORE Technologies

VisiGEN

ion torrent



Scientists – 1-2

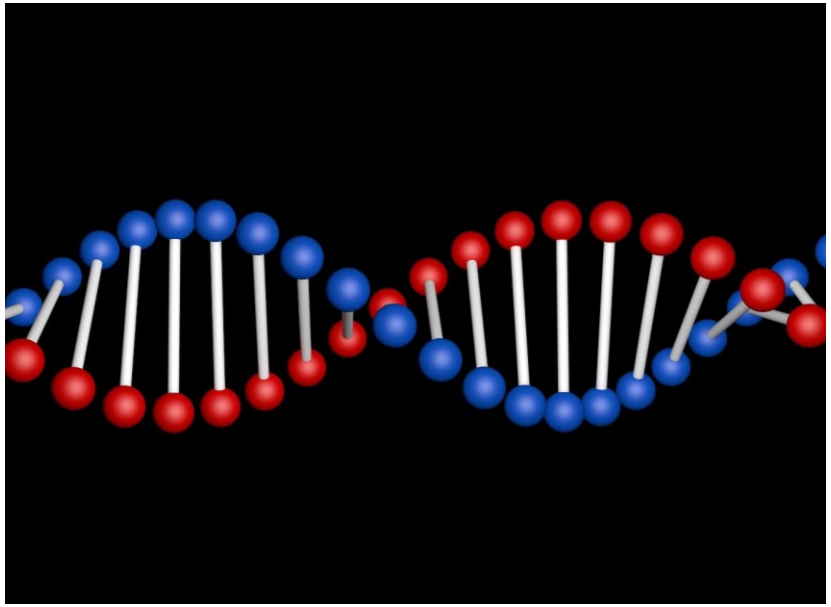
Machines - 1

Cost $1000?

Time - Hours

?

Classification of DNA Sequencing Methods



First-Generation Sequencing

Classical techniques like Sanger sequencing offer high accuracy but low throughput in DNA analysis.

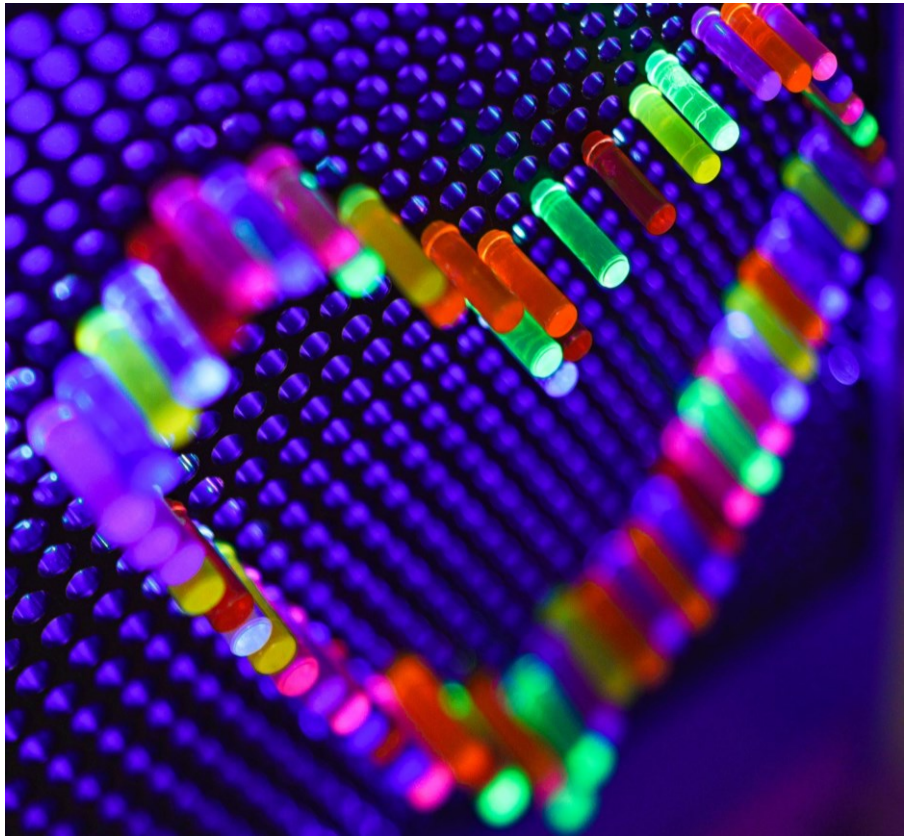
Second-Generation Sequencing

Next-generation sequencing enables massive parallel sequencing with reduced cost and time.

Third-Generation Sequencing

Single-molecule sequencing provides long read lengths to analyze complex genomic regions effectively.

First-Generation Sequencing: Sanger Method



Chain Termination Principle

Sanger sequencing uses dideoxynucleotides that terminate DNA synthesis at specific points, producing DNA fragments of varying lengths.

Fragment Separation

The DNA fragments generated are separated by size via gel electrophoresis to determine the original DNA sequence.

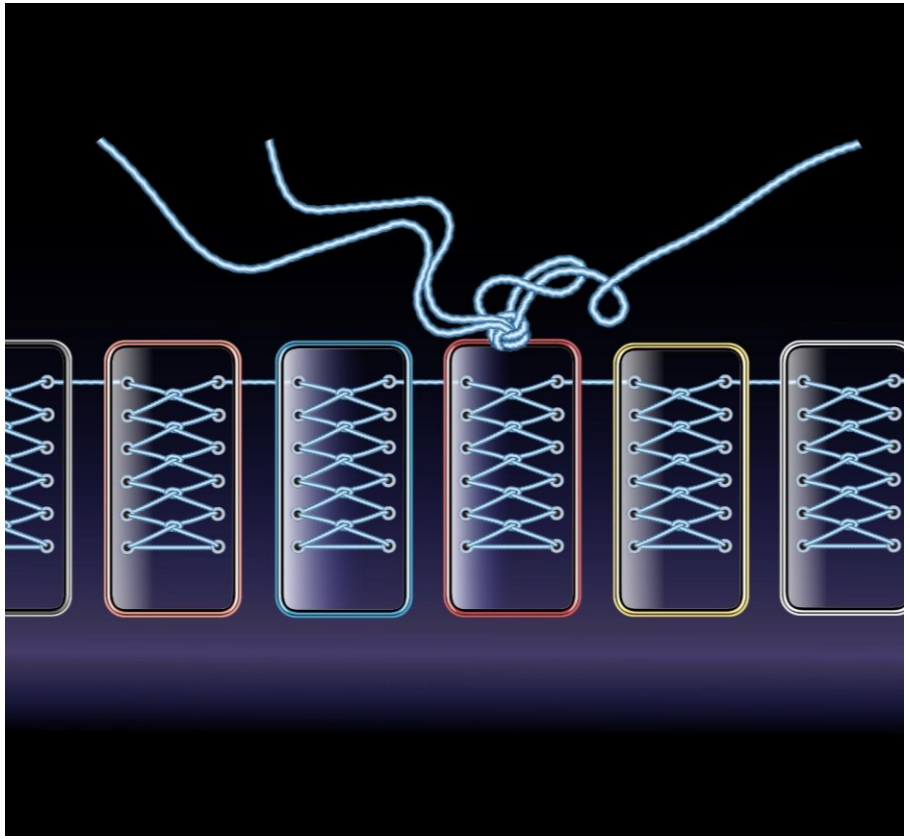
Educational Importance

Sanger sequencing demonstrates key DNA replication and enzymatic activity concepts important for biology students.

Applications and Legacy

While outdated for large genomes, Sanger sequencing remains essential for mutation detection and validating sequencing data.

Steps in Sanger Sequencing



DNA Denaturation

The DNA double helix is separated into single strands to prepare for sequencing.

Primer Binding and Extension

A primer binds the template strand, and DNA polymerase extends it by adding nucleotides.

Chain Termination

Incorporation of chain-terminating nucleotides halts DNA strand growth at varied lengths.

Fragment Separation and Detection

DNA fragments are separated by electrophoresis and detected to determine the sequence.

Second-Generation Sequencing (Next-Generation Sequencing)

Massively Parallel Sequencing

NGS sequences millions of DNA fragments simultaneously, enhancing speed and efficiency compared to traditional methods.

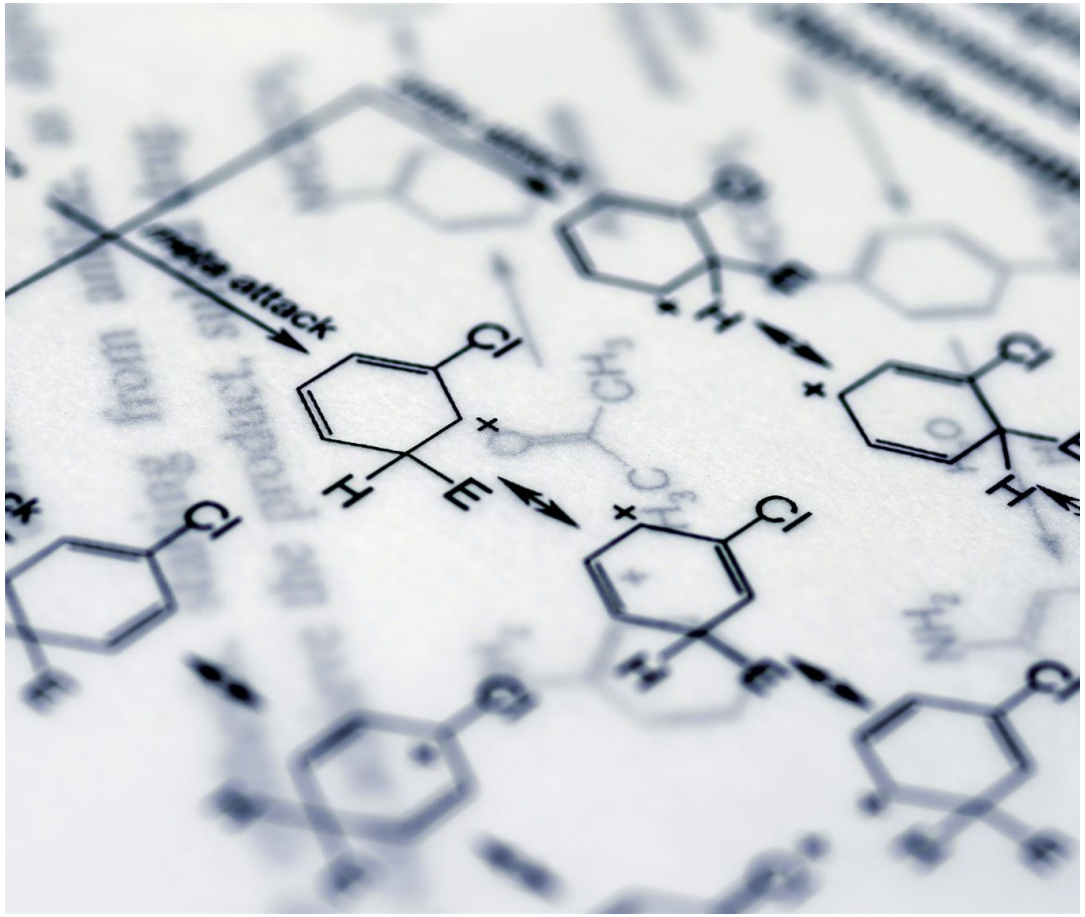
DNA Fragmentation and Amplification

DNA is fragmented into small pieces, then amplified to prepare for simultaneous sequencing on NGS platforms.

Applications in Genomics

NGS enables large-scale projects like whole-genome and transcriptome sequencing, advancing research and medicine.

Illumina Sequencing as an Example of NGS



Sequencing by Synthesis

Illumina sequencing uses fluorescently labeled nucleotides incorporated into DNA strands to determine sequences accurately.

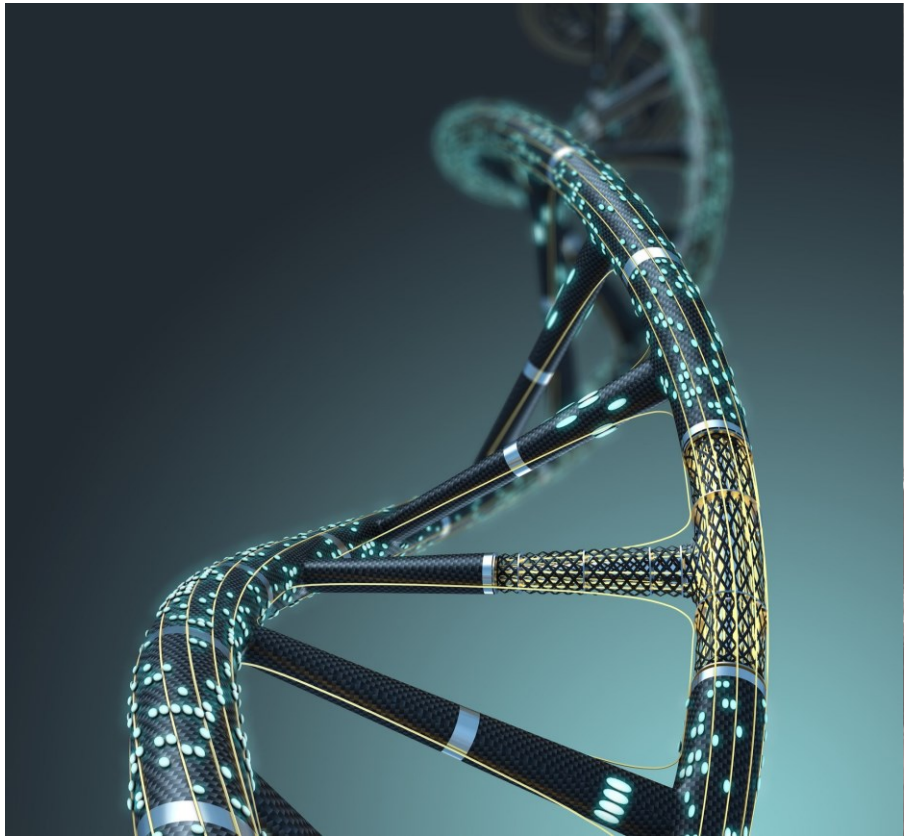
Cluster Amplification

DNA fragments attach to solid surfaces and amplify to form clusters for effective sequencing.

High Accuracy and Throughput

Illumina sequencing generates short but highly accurate reads suitable for diverse genomic applications.

Third-Generation DNA Sequencing



Direct Single-Molecule Sequencing

Third-generation sequencing reads single DNA molecules directly without amplification, enabling long read lengths.

Long Read Advantages

Longer reads simplify analysis of complex genomic regions and structural variations compared to previous technologies.

Real-Time Data and Accuracy

Reading DNA in real time introduces higher error rates, but improved data analysis enhances overall accuracy.

Applications in Genomics

Useful for genome assembly, structural variation detection, and epigenetic studies, expanding biological research scope.

Oxford Nanopore Sequencing

Nanopore Sequencing Mechanism

Biological nanopores measure electrical changes as DNA strands pass through to read nucleotide sequences.

Long Read Capability

This technology can generate extremely long DNA reads, sometimes exceeding hundreds of thousands of base pairs.

Portability and Real-Time Use

Nanopore sequencers are portable and enable real-time sequencing outside traditional labs, useful for rapid pathogen detection.

Interdisciplinary Innovation

Combines biology, physics, and computing, showcasing innovation valuable for education and modern genomics research.

Comparison of DNA Sequencing Methods

First-Generation Sequencing

Sanger sequencing offers high accuracy but low throughput, ideal for small-scale DNA studies.

Second-Generation Sequencing

Second-generation methods provide high throughput and low cost but produce short DNA reads.

Third-Generation Sequencing

Third-generation technologies generate long reads, enable real-time analysis, but with higher error rates.

Choosing the Right Method

Factors like accuracy, read length, cost, and speed determine the appropriate sequencing technology for experiments.

Human genome project on biology and technology

- Sequenced most human genes (20,000-25,000)
- Non-coding regulatory RNAs.
- The ENCODE (Encyclopedia Of DNA Elements) Project, launched by the NIH in 2003, aims to discover and understand the functional parts of the genome.
- HGP also led to the emergence of proteomics, a discipline focused on identifying and quantifying the proteins present in discrete biological compartments, such as a cellular organelle, an organ or the blood.
- Analysis of mass-spectrometry-based proteomics
- PeptideAtlas and the Trans-Proteomic Pipeline

Human genome project on biology and technology

- Our understanding of **evolution** has been transformed. Since the completion of the HGP, over 4,000 finished or quality draft genome sequences have been produced, mostly from bacterial species but including 183 eukaryotes.

- HGP drove the development of sophisticated computational and mathematical approaches to data and brought computer scientists, mathematicians, engineers and theoretical physicists together with biologists, fostering a more cross-disciplinary culture.

Human genome project on medicine

- These include the **HapMap Project** aimed at identifying haplotype blocks of common single nucleotide polymorphisms (SNPs) in different human populations and its successor, the 1000 Genomes project, an ongoing endeavor to catalogue common and rare single nucleotide and structural variation in multiple populations

- Data produced by both projects have supported smaller-scale clinical genome-wide association studies (**GWAS**), which correlate specific genetic variants with disease risk of varying statistical significance based on case–control comparisons.

- Since 2005, over 1,350 GWAS have been published.

Human genome project on medicine

- Now there are thousands of exome and whole-genome sequences (soon to be tens of thousands, and eventually millions), which have been determined with the aim of identifying disease-causing variants and, more broadly, establishing well-founded correlations between sequence variation and specific phenotypes.

- For example, the **International Cancer Genome Consortium** and **The Cancer Genome Atlas** are undertaking large-scale genomic data collection and analyses for numerous cancer types (sequencing both the normal and cancer genome for each individual patient), with a commitment to making their resources available to the research community.

Human genome project on medicine

- We predict that individual genome sequences will soon play a larger role in medical practice.

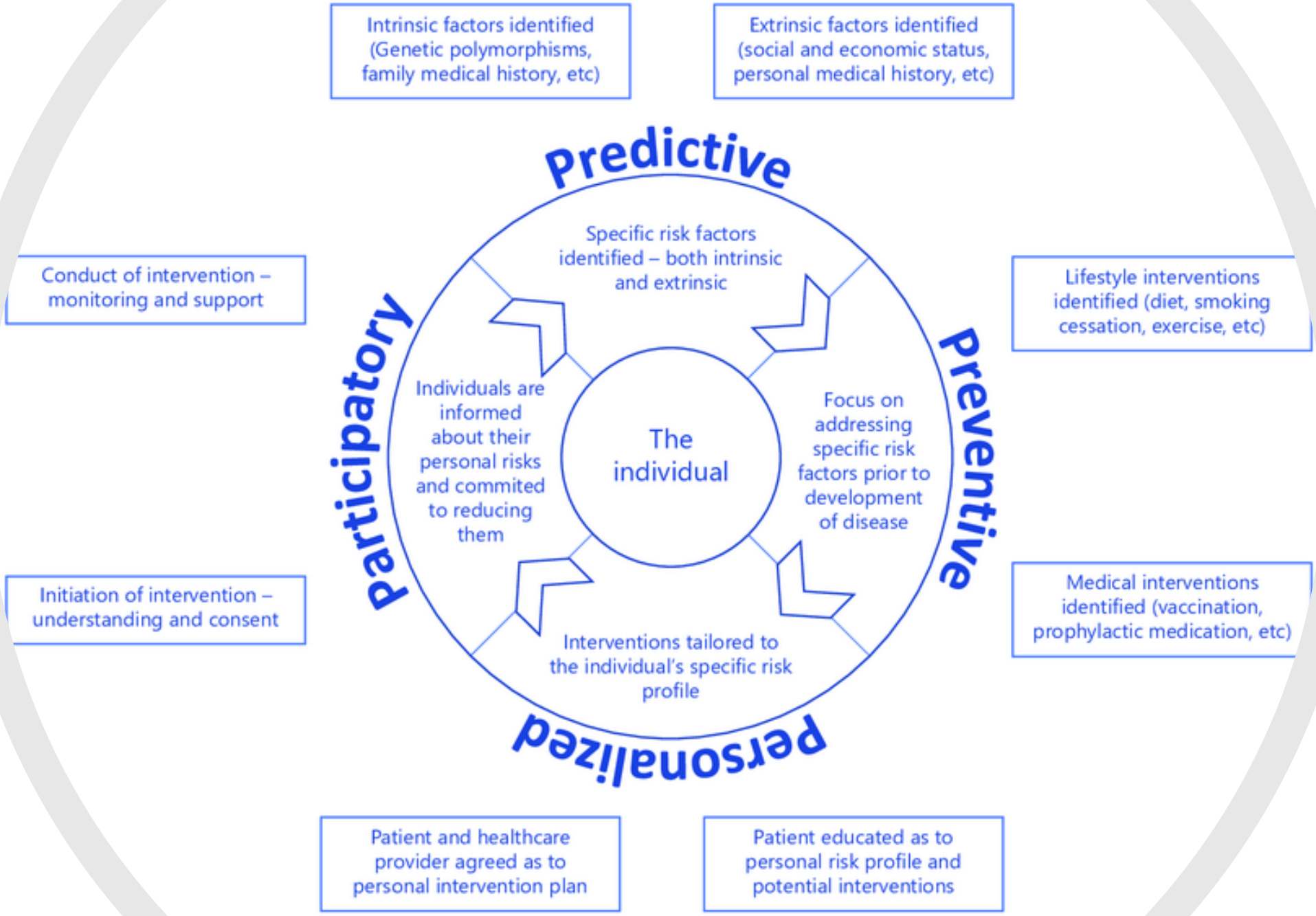
- In the ideal scenario, patients or consumers will use the information to improve their own healthcare by taking advantage of prevention or therapeutic strategies that are known to be appropriate for real or potential medical conditions suggested by their individual genome sequence.

- Physicians will need to educate themselves on how best to advise patients who bring consumer genetic data to their appointments, which may well be a common occurrence in a few years.

Human genome project on medicine

- In fact, the application of systems approaches to disease has already begun to transform our understanding of human disease and the practice of healthcare and push us towards a medicine that is predictive, preventive, personalized and participatory (P4 medicine).

- Hence, knowing how the information encoded by disease-perturbed networks changes provides insights into disease mechanisms, new approaches to diagnosis and new strategies for therapeutics.


Intrinsic factors identified (Genetic polymorphisms, family medical history, etc)
Extrinsic factors identified (social and economic status, personal medical history, etc)
Predictive
Specific risk factors identified – both intrinsic and extrinsic
Conduct of intervention – monitoring and support
Lifestyle interventions identified (diet, smoking cessation, exercise, etc)
Participatory
Individuals are informed about their personal risks and commited to reducing them
The individual
Focus on addressing specific risk factors prior to development of disease
Preventive
Initiation of intervention – understanding and consent
Medical interventions identified (vaccination, prophylactic medication, etc)
Interventions tailored to the individual's specific risk profile
Personalized
Patient and healthcare provider agreed as to personal intervention plan
Patient educated as to personal risk profile and potential interventions

Human genome project on medicine

- First, pharmacogenomics has identified more than 70 genes for which specific variants cause humans to metabolize drugs ineffectively (too fast or too slow).
- Second, there are hundreds of ‘actionable gene variants’ - variants that cause disease but whose consequences can be avoided by available medical strategies with knowledge of their presence.
- Third, in some cases, cancer-driving mutations in tumors, once identified, can be counteracted by treatments with currently available drugs.
- And last, a systems approach to blood protein diagnostics has generated powerful new diagnostic panels for human diseases such as hepatitis and lung cancer.

Human genome project on medicine

- These latter examples portend a revolution in blood diagnostics that will lead to early detection of disease, the ability to follow disease progression and responses to treatment, and the ability to stratify a disease type (for instance, breast cancer) into its different subtypes for proper impedance match against effective drugs.

- In the future when all patients will be surrounded by a virtual cloud of billions of data points, and when we will have the analytical tools to reduce this enormous data dimensionality to simple hypotheses to optimize wellness and minimize disease for each individual.

10K genomes project (UK10K)

- The UK10K project was a 2010–2013 collaboration among several institutions including Bristol University, King's College London, the Medical Research Council, the UK Department of Health and the Wellcome Trust Sanger Institute (the biggest funding partner).
- The principal goal was to identify novel disease-causing genes linked to low-frequency and rare genetic changes.
- DNA samples were collected from close to 5500 individuals selected from the participants in 11 different disease-specific studies and whole-exome sequences (WES) obtained to 72x depth. 4000 control samples were included from two cohort studies in the UK
- UK10K has implemented mechanisms to report back to participants any clinically significant findings that might arise during the lifetime of the project.

100K genomes project

- In late 2012 Genomics England launched 100K genomes project (100KGP) with the main goal of sequencing 100 000 whole genomes from about 70 000 individuals by 2017.
- Participants would be patients with a focus on those with a rare disease or cancer, plus selected family members, due to the fact that both cancer and rare diseases are strongly linked to changes in the genome.
- The aim is to sequence 50 000 genomes from cancer (25 000 patients with two samples each, one from tumor and one from normal tissue), and 50 000 genomes from rare diseases (roughly 17 000 patients with three samples taken, one from the affected person plus two from blood relatives).
- There are four overarching goals: to create an ethical and transparent program based on consent, to set up a genomic medicine service bringing benefit to patients, to promote new scientific and medical discovery and to develop a strong UK genomics industry. Sampling is ongoing in England, and discussions about potential involvement are taking place with Scotland, Wales and Northern Ireland.

Genome Asia 100K

- Genome Asia 100K (GA100K) is a project delivered by a private consortium constituted by the companies Macrogen and MedGenome.
- Asian people are under-represented in the current genome databases, which are largely biased toward genomes from Caucasian people, despite the fact that Asian people account for more than 40% of the world population.
- Therefore, the main goal is to sequence and analyze a total of 100 000 Asian individual genomes to drive precision medicine.
- The first stage of this project is to investigate ethnic stratification by sequencing 10 000 Asian individuals, since there is evidence that mutations appear at different frequencies in distinct ethnic groups. A second stage is to sequence the remaining 90 000 individuals, together with the collection of phenotype and clinical data.
- This information will enable a deep analysis of the Asian population in health and disease, of relevance to rare and inherited Mendelian diseases, and to complex diseases such as cancer, diabetes and neurological disorders. Longitudinal studies are also planned.

The precision medicine initiative

- In 2015, the then president of the USA, Barack Obama, made the precision medicine initiative (PMI) announcement in his State of the Union Address.
- This initiative has two major components: a short-term focus on cancer and a long-term focus on acquiring better knowledge about health and disease.
- Specifically, the ultimate objectives of the PMI are as follows: getting more and better treatments for cancer, the creation of a voluntary national research cohort of at least 1 million people, a strong enhancement of the protection of privacy, improving regulatory mechanisms and forging public-private partnerships.
- To accomplish such goals, $215 million were invested in 2016 by the NIH, the Food and Drug Administration (FDA) and the Office of the National Coordinator for Health Information Technology (ONC).

China precision medicine initiative

- On January 2016 the Chinese Academy of Sciences (CAS) announced the Chinese Precision Medicine Initiative (CPMI), which will have received government funding of $9.2 billion by 2030.
- The Chinese government has confirmed that precision medicine will be part of its Five Year Plan for 2016–2020.
- Official announcements of the plan were made during the National People's Congress. This project will be led by the BGI which will manage the research work from several institutes collecting genetic information from 4000 volunteers in the first four years.
- Deep data mining will be performed on half of these volunteers, including whole-genome sequence analysis, elaboration of genome sequence-related health profiles, evaluation of risk of chronic diseases associated to genetic factors and prediction of patient reactions to drugs.

- Human Genome Project will allow scientists to develop new ways of treating or diagnosing illnesses, especially genetic disorders and cancer.

- Genomic medicine is an emerging medical discipline that involves using genomic information about an individual as part of their clinical care (e.g. for diagnostic or therapeutic decision-making) and the health outcomes and policy implications of that clinical use.

- Already, genomic medicine is making an impact in the fields of oncology, pharmacology, rare and undiagnosed diseases, and infectious disease.

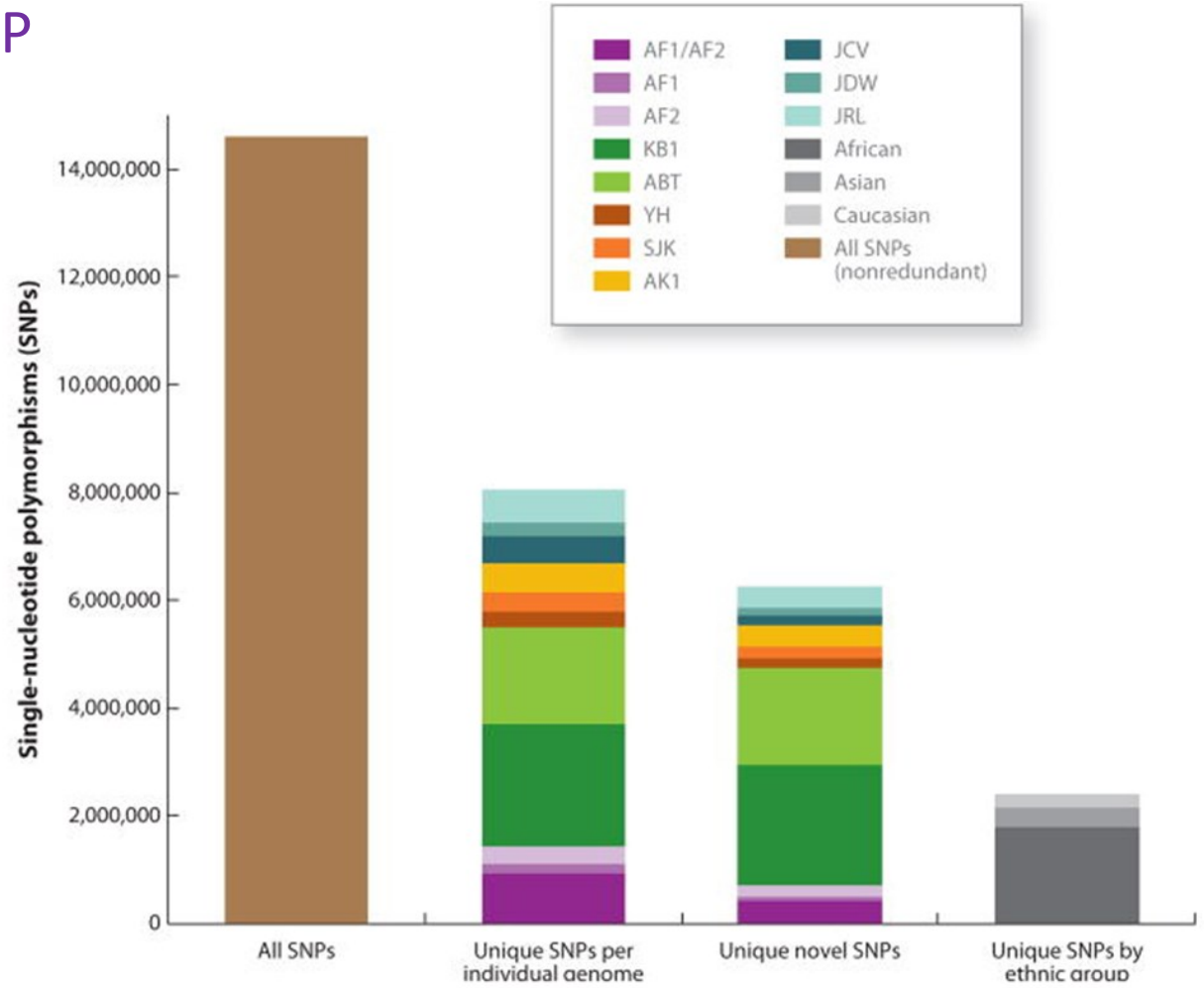
Comparison of sequenced personal human genomes

Individual	Ploidy	Technology	Av Depth	Total SNPs [M]	Known SNPs [M] (%)	Novel SNPs [M] (%)	Heterozygous SNPs [M] (%)	Homozygous SNPs [M] (%)	cSNPs	nsSNPs	InDels	CNVs (≥100 bp)
Venter	2n	Sanger	7.5×	3.21	2.80 (87.22%)	0.41 (12.77%)	1.76 (54.85%)	1.45 (45.15%)	21,152	6,114	214,691	6,485
Watson	2n	Roche 454	7.4×	3.32	2.71 (81.73%)	0.61 (18.27%)	1.67 (50.53%)	1.64 (49.47%)	22,041	10,659	222,718	1,674
Chinese (YH)	2n	Illumina	36.0×	3.07	2.65 (87.13%)	0.41 (12.87%)	1.72 (56.03%)	1.35 (43.97%)	15,759	7,062	135,262	2,682
African (NA18507)*	2n	Illumina	40.6×	3.61	2.72 (75.50%)	0.88 (24.50%)	2.28 (63.21%)	1.32 (36.79%)	26,140	5,361	404,416	8,470
African (NA18507)*	2n	AB SOLiD	17.9×	3.86	3.13 (81.00%)	0.73 (19.00%)	2.33 (60.30%)	1.53 (39.70%)	68,624	9,902	226,529	6,714
Korean (SJK)	2n	Illumina	28.9×	3.43	3.01 (87.79%)	0.42 (12.21%)	2.00 (58.21%)	1.43 (41.79%)	27,118	9,334	342,965	3,303
Korean (AK1)	2n	Illumina	27.8×	3.45	2.86 (83.30%)	0.59 (16.70%)	2.11 (61.11%)	1.34 (38.89%)	21,606	10,162	170,202	414
Khoisan (KB1)	2n	Roche 454	10.2×	4.05	3.31 (81.65%)	0.74 (18.35%)	2.39 (59.00%)	1.66 (41.00%)	22,119	na	463,788	na
D. Tutu (ABT)	2n	AB SOLiD	30.0×	3.62	3.21 (88.61%)	0.41 (11.39%)	2.17 (60.00%)	1.44 (40.00%)	17,342	na	3,395	na
Lupski	2n	AB SOLiD	29.6×	3.42	2.85 (83.58%)	0.56 (16.42%)	2.00 (58.72%)	1.41 (41.28%)	18,406	9,069	na	530

*Same HapMap sample was independently sequenced and reported using two different technologies.

Abbreviations: cSNPs, coding SNPs; nsSNPs, nonsynonymous SNPs; CNVs, copy-number variants; na, data not available.

SNP



Comparison of single nucleotide polymorphisms (SNPs) in 10 personal genomes. All SNPs in any of 10 sequenced personal genomes were compared with the other 9 genomes. Altogether, the 10 genomes contribute 14,608,404 nonredundant SNPs (first bar). The second bar pictures all SNPs that are unique to each of the personal genomes; the third bar represents all the SNPs that are unique in a given personal genome but also novel; the fourth bar shows the SNPs shared by individuals of the same ethnic group. Abbreviations: AF1, NA18507(1) Illumina; AF2, NA18507(2) SOLiD; KB1, Khoisan genome; ABT, Archbishop Desmond Tutu; YH, Chinese genome; SJK, Korean genome 1; AK1, Korean genome 2; JCV, J. Craig Venter; JDW, James D. Watson; JRL, James R. Lupski.

A comparison of the weaknesses and strengths of whole-genome sequencing (WGS) and exome sequencing approaches for disease-gene identification. Abbreviations: CNVs, copy-number variants; SNVs, simple nucleotide variants.

	Whole-genome sequencing (WGS)	Exome sequencing
Cost	Still costly, but decreasing rapidly	Reduced cost is a tenth to a third of WGS
Technical	No capture step, automatable	Capture step, technical bias
Variation	Uncovers all genetic and and genomic variation (SNVs and CNVs) Discovery of functional coding and noncoding variation ~3.5 million variants	Focuses on ~1% of the genome Limited to coding and splice-site variants in annotated genes ~20,000 variants
Disease	Suitable for mendelian and complex trait gene identification, as well as sporadic phenotypes caused by *de novo* SNVs or CNVs	Good for highly penetrant mendelian disease gene identification

Completing The Puzzle

- Although the human genome has been completely read by DNA sequencing, it is not yet fully understood. Having the complete sequence of the human genome is similar to having all the pages of a manual needed to make the human body. The challenge now is to determine how to read the contents of these pages and understand how all of these many, complex parts work together in human health and disease.

- As a result from the Human Genome Project, there are now more than 2,000 genetic tests for human conditions with 350 biotechnology-based products currently in clinical trials. These tests enable patients to learn their genetic risks for disease and help healthcare professionals diagnose disease.

Future hopes

- It is hoped that more and more medicines can be developed now that the sequence of the human genome is known.

- Being able to target diseases with specific medicines, and using gene therapy to treat diseases on a patient-by-patient basis, will revolutionize modern medicine.

- Patients can also respond to drugs differently. Sometimes one disease can be caused by a variety of different mutations or risk factors.

- Pharmacogenetics is the study of an individual's genome to predict how they will respond to a drug treatment. This type of medicine will also help to make sure that drugs are only used if the doctors know they are going to work. This means the patient is more likely to respond to the drugs and also reduces the use of ineffective drugs, saving limited resources.

Risk factors and causal mechanisms/ Correlation

- Scientists look for patterns in data and links between variables. If links or patterns are found this is called correlation as one variable changes, so does the other.
- Research has established links between cancer and various lifestyle factors, chemicals produced in the body, or that enter the human body, and chemicals in the environment.
- Scientists have established several causal mechanisms for risk factors.
- Below is an example of how risk factors have been found. This has come from studies on smoking cigarettes and lung cancer.
- Historically, in the USA, a pattern can be seen between the number of cigarettes smoked and the number of lung cancer deaths.

Video-
https://www.youtube.com/watch?v=C4dT3JWXCDI

https://www.youtube.com/watch?v=MvuYATh7Y74

https://www.youtube.com/watch?v=qOW5e4BgEa4

Thank you